

Task 5 - CO323

E/21/019

Part A — Single-Hop Routing

Network Topology and Addressing

Three PCs were connected in a linear topology. The middle PC was configured with two interfaces, each belonging to a different subnet, allowing it to function as a router between the networks. The addressing plan is shown above.

Device	Interface	IP Address	Subnet Mask	Role
PC0 (left)	NIC	192.168.1.10	255.255.255.0	End host
PC-Router (middle)	NIC 1	192.168.1.1	255.255.255.0	Gateway — left subnet
PC-Router (middle)	NIC 2	192.168.2.1	255.255.255.0	Gateway — right subnet
PC1 (right)	NIC	192.168.2.10	255.255.255.0	End host

Each end host was configured with an IP address and a default gateway set to the closest interface of the middle PC. The middle PC was assigned a separate IP address for each interface. Static routing was configured on the middle PC to allow traffic to pass between the two subnets. The setup was verified by successfully sending ICMP echo (ping) requests from PC0 to PC1 and receiving responses.

08:51:07.284004	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=19/4864, ttl=128 (reply in 2)
08:51:07.285429	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=19/4864, ttl=127 (request in 1)
08:51:08.293124	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=20/5120, ttl=128 (reply in 4)
08:51:08.294712	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=20/5120, ttl=127 (request in 3)
08:51:09.323477	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=21/5376, ttl=128 (reply in 6)
08:51:09.325128	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=21/5376, ttl=127 (request in 5)
08:51:10.356231	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=22/5632, ttl=128 (reply in 8)
08:51:10.357903	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=22/5632, ttl=127 (request in 7)

Figure 1. IP configuration on PC0 (left end host).

1 08:51:39.795368	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) request	id=0x0001, seq=13/3328, ttl=128 (reply in 2)
2 08:51:39.797031	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=13/3328, ttl=127 (request in 1)
3 08:51:40.799522	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) request	id=0x0001, seq=14/3584, ttl=128 (reply in 4)
4 08:51:40.801305	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=14/3584, ttl=127 (request in 3)
5 08:51:41.814089	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) request	id=0x0001, seq=15/3840, ttl=128 (reply in 6)
6 08:51:41.815820	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=15/3840, ttl=127 (request in 5)
7 08:51:42.821355	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) request	id=0x0001, seq=16/4096, ttl=128 (reply in 8)
8 08:51:42.823239	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=16/4096, ttl=127 (request in 7)
9 08:51:44.554631	CompalInform_7d:a4:...	EdimaxTechno_5f:9c:...	ARP	42 Who has 192.168.2.1? Tell 192.168.2.10	
10 08:51:44.555408	EdimaxTechno_5f:9c:...	CompalInform_7d:a4:...	ARP	60 192.168.2.1 is at 08:50:fc:5f:9c:73	

Figure 2. IP configuration on PC1 (right end host).

2 03:21:44.743363	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=19/4864, ttl=127 (reply in 3)
3 03:21:44.744322	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=19/4864, ttl=128 (request in 2)
4 03:21:45.752660	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=20/5120, ttl=127 (reply in 5)
5 03:21:45.753522	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=20/5120, ttl=128 (request in 4)
6 03:21:46.782979	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=21/5376, ttl=127 (reply in 7)
7 03:21:46.784018	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=21/5376, ttl=128 (request in 6)
8 03:21:47.815772	192.168.1.10	192.168.2.10	ICMP	74 Echo (ping) request	id=0x0001, seq=22/5632, ttl=127 (reply in 9)
9 03:21:47.816808	192.168.2.10	192.168.1.10	ICMP	74 Echo (ping) reply	id=0x0001, seq=22/5632, ttl=128 (request in 8)

Figure 3. Interface and route configuration on the middle routing PC.

38	03:22:19.471126	192.168.2.10	192.168.1.10	ICMP	74	Echo (ping) request	id=0x0001, seq=13/3328, ttl=128 (reply in 39)
39	03:22:19.472112	192.168.1.10	192.168.2.10	ICMP	74	Echo (ping) reply	id=0x0001, seq=13/3328, ttl=127 (request in 38)
40	03:22:20.475259	192.168.2.10	192.168.1.10	ICMP	74	Echo (ping) request	id=0x0001, seq=14/3584, ttl=128 (reply in 41)
41	03:22:20.476318	192.168.1.10	192.168.2.10	ICMP	74	Echo (ping) reply	id=0x0001, seq=14/3584, ttl=127 (request in 40)
42	03:22:21.489808	192.168.2.10	192.168.1.10	ICMP	74	Echo (ping) request	id=0x0001, seq=15/3840, ttl=128 (reply in 43)
43	03:22:21.490820	192.168.1.10	192.168.2.10	ICMP	74	Echo (ping) reply	id=0x0001, seq=15/3840, ttl=127 (request in 42)
44	03:22:22.497097	192.168.2.10	192.168.1.10	ICMP	74	Echo (ping) request	id=0x0001, seq=16/4096, ttl=128 (reply in 45)
45	03:22:22.498148	192.168.1.10	192.168.2.10	ICMP	74	Echo (ping) reply	id=0x0001, seq=16/4096, ttl=127 (request in 44)
46	03:22:24.230317	CompalInform_7d:a4:...	EdimaxTechno_5f:9c:...	ARP	60	Who has 192.168.2.1? Tell 192.168.2.10	
47	03:22:24.230346	EdimaxTechno_5f:9c:...	CompalInform_7d:a4:...	ARP	42	192.168.2.1 is at 00:50:fc:5f:9c:73	
48	03:22:45.753138	192.168.2.1	192.168.2.255	BROWSER	243	Local Master Announcement CE-559, Workstation, Server, NT Workstation, Potential Browser, Master Browser	
49	03:22:45.826312	192.168.2.1	192.168.2.255	BROWSER	249	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum	
50	03:23:02.819698	CompalInform_7d:a4:...	Broadcast	ARP	60	Who has 192.168.2.1? Tell 192.168.2.10	
51	03:23:02.819723	EdimaxTechno_5f:9c:...	CompalInform_7d:a4:...	ARP	42	192.168.2.1 is at 00:50:fc:5f:9c:73	
52	03:23:02.826455	CompalInform_7d:a4:...	Broadcast	ARP	60	Who has 10.118.121.42? Tell 192.168.2.10	
53	03:23:03.735300	CompalInform_7d:a4:...	Broadcast	ARP	60	Who has 10.118.121.42? Tell 192.168.2.10	
54	03:23:04.732180	CompalInform_7d:a4:...	Broadcast	ARP	60	Who has 10.118.121.42? Tell 192.168.2.10	

Figure 4. Successful ping from PC0 to PC1 after routing configuration.

Results

Test	Observed Result
PC0 → 192.168.1.1 (local gateway)	Success
PC1 → 192.168.2.1 (local gateway)	Success
PC0 → PC1 before routing	Fail
PC0 → PC1 after routing	Success
PC1 → PC0 after routing	Success

Part B — Multi-Hop Linear Routing

Network Topology and Addressing

Four nodes (A, B, C, and D) were connected using three separate /24 subnets. Nodes B and C were configured with two network interfaces each, allowing them to operate as intermediate routers between the subnets.

Device	Interface	Address	Subnet Mask	Role
A	eth0	10.10.0.2	255.255.255.0	End host
B	eth0	10.10.0.1	255.255.255.0	Router — facing A
B	eth1	10.10.1.1	255.255.255.0	Router — facing C
C	eth0	10.10.1.2	255.255.255.0	Router — facing B
C	eth1	10.10.2.1	255.255.255.0	Router — facing D
D	eth0	10.10.2.2	255.255.255.0	End host

IP forwarding was enabled on both B and C. Static routes were configured so that B forwards traffic for the 10.10.2.0/24 network to C, and C forwards traffic for the 10.10.0.0/24 network back to B. Host A was configured to send traffic to B as its gateway, while host D used C as its gateway. End-to-end connectivity was verified by successfully pinging from A to D, and the routing tables on B and C were checked to ensure the routes were correctly installed.

```

C:\Windows\system32>route print
=====
Interface List
70...00 e0 4c 69 05 62 .....Realtek USB GbE Family Controller
22...2e 33 7a fa ee f9 .....Microsoft Wi-Fi Direct Virtual Adapter
17...2e 33 7a fa e6 f9 .....Microsoft Wi-Fi Direct Virtual Adapter #2
21...74 e6 e2 48 ff 57 .....Realtek PCIe FE Family Controller
12...00 ff 88 b4 ff c9 .....Kaspersky Security Data Escort Adapter
16...00 ff ac c6 39 1f .....TAP-Windows Adapter V9
20...2c 33 7a fa ee f9 .....Dell Wireless 1704 802.11b/g/n (2.4GHz)
6...2c 33 7a fa ee fa .....Bluetooth Device (Personal Area Network) #2
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.118.121.42    10.118.121.186   55
10.0.0.0                   255.255.255.0    10.0.1.1         10.0.1.2         36
10.0.1.0                   255.255.255.0    On-link          10.0.1.2         291
10.0.1.2                   255.255.255.255  On-link          10.0.1.2         291
10.0.1.255                 255.255.255.255  On-link          10.0.1.2         291
10.0.2.0                   255.255.255.0    On-link          10.0.2.1         281
10.0.2.1                   255.255.255.255  On-link          10.0.2.1         281
10.0.2.255                 255.255.255.255  On-link          10.0.2.1         281
10.118.121.0               255.255.255.0    On-link          10.118.121.186   311
10.118.121.186             255.255.255.255  On-link          10.118.121.186   311
10.118.121.255             255.255.255.255  On-link          10.118.121.186   311
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.2.0                255.255.255.0    192.168.1.1     10.0.1.2         36
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          10.0.1.2         291
224.0.0.0                  240.0.0.0        On-link          10.118.121.186   311
224.0.0.0                  240.0.0.0        On-link          10.0.2.1         281
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          10.0.1.2         291
255.255.255.255            255.255.255.255  On-link          10.118.121.186   311
255.255.255.255            255.255.255.255  On-link          10.0.2.1         281
=====

```

Figure 5. Routing table on node B showing directly connected and static routes.

```

=====
Persistent Routes:
Network Address            Netmask          Gateway Address  Metric
192.168.2.0                255.255.255.0    192.168.1.1     1
10.0.0.0                   255.255.255.0    10.0.1.1        1
=====

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
20      71 :::/0                          fe80::58f4:85ff:fe41:7962
1       331 ::1/128                        On-link
20      71 2407:c00:6007:b8a2::/64         On-link
20      311 2407:c00:6007:b8a2:3d7f:d95:e191:86da/128
                                           On-link
20      311 2407:c00:6007:b8a2:a8b1:92a5:f7dd:63f1/128
                                           On-link
21      291 fe80::/64                       On-link
20      311 fe80::/64                       On-link
70      281 fe80::/64                       On-link
70      281 fe80::5129:a43a:1492:3e81/128
                                           On-link
21      291 fe80::5284:f19d:8129:c93a/128
                                           On-link
20      311 fe80::c60a:811c:7932:a990/128
                                           On-link
1       331 ff00::/8                         On-link
21      291 ff00::/8                         On-link
20      311 ff00::/8                         On-link
70      281 ff00::/8                         On-link
=====
Persistent Routes:
None
=====

```

Figure 6. Routing table on node C showing directly connected and static routes.

```

C:\Windows\system32>route print
=====
Interface List
12...74 46 a0 91 c0 86 .....Intel(R) 82579LM Gigabit Network Connection
10...00 50 fc 5f 9c 73 .....Realtek RTL8139/810x Family Fast Ethernet NIC
1.....Software Loopback Interface 1
8...00 00 00 00 00 00 e0 Microsoft ISATAP Adapter
4...00 00 00 00 00 00 e0 Microsoft ISATAP Adapter #2
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
10.0.0.0                    255.255.255.0    On-link           10.0.0.1          276
10.0.0.1                    255.255.255.255  On-link           10.0.0.1          276
10.0.0.255                  255.255.255.255  On-link           10.0.0.1          276
10.0.1.0                    255.255.255.0    On-link           10.0.1.1          276
10.0.1.1                    255.255.255.255  On-link           10.0.1.1          276
10.0.1.255                  255.255.255.255  On-link           10.0.1.1          276
127.0.0.0                   255.0.0.0        On-link           127.0.0.1         306
127.0.0.1                   255.255.255.255  On-link           127.0.0.1         306
127.255.255.255             255.255.255.255  On-link           127.0.0.1         306
224.0.0.0                   240.0.0.0        On-link           127.0.0.1         306
224.0.0.0                   240.0.0.0        On-link           10.0.1.1          276
224.0.0.0                   240.0.0.0        On-link           10.0.0.1          276
255.255.255.255             255.255.255.255  On-link           127.0.0.1         306
255.255.255.255             255.255.255.255  On-link           10.0.1.1          276
255.255.255.255             255.255.255.255  On-link           10.0.0.1          276
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
1       306 ::1/128                      On-link
12      276 fe80::/64                      On-link
10      276 fe80::/64                      On-link
12      276 fe80::c8c9:cdc:d73:4945/128    On-link
10      276 fe80::cded:ea62:881b:1a6c/128    On-link
1       306 ff00::/8                      On-link
12      276 ff00::/8                      On-link
10      276 ff00::/8                      On-link
=====
Persistent Routes:
None

```

Figure 7. Full routing configuration on nodes B and C.

```

PS C:\WINDOWS\system32> ping 10.10.2.0

Pinging 10.10.2.0 with 32 bytes of data:
Reply from 10.10.2.0: bytes=32 time<1ms TTL=127
Reply from 10.10.2.0: bytes=32 time<1ms TTL=127
Reply from 10.10.2.0: bytes=32 time<1ms TTL=127
Reply from 10.10.2.0: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.2.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Figure 8. Successful ping from A to D across three subnets.

Results

Test	Observed Result
A → B (direct)	Success
B → C (direct)	Success
C → D (direct)	Success
A → D before static routing	Fail
A → D after static routing	Success
D → A after static routing	Success

Part C — Ring Topology with Dual Paths

Network Topology and Addressing

Six PCs (A to F) were arranged in a ring topology, with each PC using two Ethernet interfaces to connect to its neighboring nodes. This setup formed six separate /24 subnets and provided two distinct physical paths between any non-adjacent pair of nodes.

PC	Link	IP Address	Subnet
A	A — B	10.10.0.1	10.10.0.0/24
A	A — F	10.10.5.2	10.10.5.0/24
B	B — A	10.10.0.2	10.10.0.0/24
B	B — C	10.10.1.1	10.10.1.0/24
C	C — B	10.10.1.2	10.10.1.0/24
C	C — D	10.10.2.1	10.10.2.0/24
D	D — C	10.10.2.2	10.10.2.0/24
D	D — E	10.10.3.1	10.10.3.0/24
E	E — D	10.10.3.2	10.10.3.0/24
E	E — F	10.10.4.1	10.10.4.0/24
F	F — E	10.10.4.2	10.10.4.0/24
F	F — A	10.10.5.1	10.10.5.0/24

Pre-Configuration Checks

Because the lab machines are shared, each PC was inspected before any configuration was applied. The following commands were run on every node:

```
hostname
ip -br link
ip -br addr
```

```
ip route
```

Any pre-existing addresses and routes on the lab interfaces were removed to avoid conflicts:

```
sudo ip addr flush dev <interface>
sudo ip route flush dev <interface>
sudo ip link set <interface> up
```

Enabling IP Forwarding

IPv4 forwarding was activated on all six PCs before any routing tests were run:

```
sudo sysctl -w net.ipv4.ip_forward=1
```

Direct-Link Verification

Before adding static routes, each directly connected pair of neighboring PCs was tested using ping. This helped identify any physical or IP addressing issues independently of routing configuration problems.

PC A:

```
ceadmin@ce-558:~$ ping -c 4 10.10.0.2
PING 10.10.0.2 (10.10.0.2) 56(84) bytes of data.
64 bytes from 10.10.0.2: icmp_seq=1 ttl=64 time=0.324 ms
64 bytes from 10.10.0.2: icmp_seq=2 ttl=64 time=0.258 ms
64 bytes from 10.10.0.2: icmp_seq=3 ttl=64 time=0.236 ms
64 bytes from 10.10.0.2: icmp_seq=4 ttl=64 time=0.273 ms

--- 10.10.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.236/0.272/0.324/0.032 ms
ceadmin@ce-558:~$ ping -c 4 10.10.5.1
PING 10.10.5.1 (10.10.5.1) 56(84) bytes of data.
64 bytes from 10.10.5.1: icmp_seq=1 ttl=64 time=0.412 ms
64 bytes from 10.10.5.1: icmp_seq=2 ttl=64 time=0.228 ms
64 bytes from 10.10.5.1: icmp_seq=3 ttl=64 time=0.193 ms
64 bytes from 10.10.5.1: icmp_seq=4 ttl=64 time=0.245 ms

--- 10.10.5.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3072ms
rtt min/avg/max/mdev = 0.193/0.269/0.412/0.084 ms
```

Figure 9. Direct-link ping test on PC A.

PC B:

```

ceadmin@ce-570:~$ ping -c 4 10.10.0.1
PING 10.10.0.1 (10.10.0.1) 56(84) bytes of data.
64 bytes from 10.10.0.1: icmp_seq=1 ttl=64 time=0.249 ms
64 bytes from 10.10.0.1: icmp_seq=2 ttl=64 time=0.255 ms
64 bytes from 10.10.0.1: icmp_seq=3 ttl=64 time=0.257 ms
^[[A64 bytes from 10.10.0.1: icmp_seq=4 ttl=64 time=0.282 ms

--- 10.10.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3077ms
rtt min/avg/max/mdev = 0.249/0.260/0.282/0.012 ms
ceadmin@ce-570:~$ ping -c 4 10.10.1.2
PING 10.10.1.2 (10.10.1.2) 56(84) bytes of data.
64 bytes from 10.10.1.2: icmp_seq=1 ttl=64 time=0.174 ms
64 bytes from 10.10.1.2: icmp_seq=2 ttl=64 time=0.243 ms
64 bytes from 10.10.1.2: icmp_seq=3 ttl=64 time=0.189 ms
64 bytes from 10.10.1.2: icmp_seq=4 ttl=64 time=0.280 ms

--- 10.10.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.174/0.221/0.280/0.042 ms
ceadmin@ce-570:~$

```

Figure 10. Direct-link ping test on PC B.

PC C:

```

ceadmin@ceadmin-OptiPlex-3020:~$ ping -c 2 10.10.2.2
PING 10.10.2.2 (10.10.2.2) 56(84) bytes of data.
64 bytes from 10.10.2.2: icmp_seq=1 ttl=64 time=0.190 ms
64 bytes from 10.10.2.2: icmp_seq=2 ttl=64 time=0.221 ms

--- 10.10.2.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.190/0.205/0.221/0.015 ms
ceadmin@ceadmin-OptiPlex-3020:~$ ping -c 2 10.10.1.1
PING 10.10.1.1 (10.10.1.1) 56(84) bytes of data.
64 bytes from 10.10.1.1: icmp_seq=1 ttl=64 time=0.255 ms
64 bytes from 10.10.1.1: icmp_seq=2 ttl=64 time=0.238 ms

--- 10.10.1.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1011ms
rtt min/avg/max/mdev = 0.238/0.246/0.255/0.008 ms

```

Figure 11. Direct-link ping test on PC C.

PC D:

```

ceadmin@ce-551:~$ ping -c2 10.10.3.2
PING 10.10.3.2 (10.10.3.2) 56(84) bytes of data.
64 bytes from 10.10.3.2: icmp_seq=1 ttl=64 time=0.497 ms
64 bytes from 10.10.3.2: icmp_seq=2 ttl=64 time=0.220 ms

--- 10.10.3.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1009ms
rtt min/avg/max/mdev = 0.220/0.358/0.497/0.138 ms
ceadmin@ce-551:~$ ping -c2 10.10.2.1
PING 10.10.2.1 (10.10.2.1) 56(84) bytes of data.
64 bytes from 10.10.2.1: icmp_seq=1 ttl=64 time=0.360 ms
64 bytes from 10.10.2.1: icmp_seq=2 ttl=64 time=0.253 ms

--- 10.10.2.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1030ms
rtt min/avg/max/mdev = 0.253/0.306/0.360/0.053 ms

```

Figure 12. Direct-link ping test on PC D.

PC E:

```

ceadmin@ce-553:~$ ping -c2 10.10.4.2
PING 10.10.4.2 (10.10.4.2) 56(84) bytes of data.
64 bytes from 10.10.4.2: icmp_seq=1 ttl=64 time=1.70 ms
64 bytes from 10.10.4.2: icmp_seq=2 ttl=64 time=0.973 ms

--- 10.10.4.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.973/1.335/1.698/0.362 ms
ceadmin@ce-553:~$ ping -c2 10.10.3.1
PING 10.10.3.1 (10.10.3.1) 56(84) bytes of data.
64 bytes from 10.10.3.1: icmp_seq=1 ttl=64 time=0.244 ms
64 bytes from 10.10.3.1: icmp_seq=2 ttl=64 time=0.277 ms

--- 10.10.3.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1019ms
rtt min/avg/max/mdev = 0.244/0.260/0.277/0.016 ms

```

Figure 13. Direct-link ping test on PC E.

PC F:

```

ceadmin@ceadmin-HP-Compaq-Pro-6300-MT:~$ ping -c 4 10.10.5.2
PING 10.10.5.2 (10.10.5.2) 56(84) bytes of data.
64 bytes from 10.10.5.2: icmp_seq=1 ttl=64 time=0.310 ms
64 bytes from 10.10.5.2: icmp_seq=2 ttl=64 time=0.318 ms
64 bytes from 10.10.5.2: icmp_seq=3 ttl=64 time=0.275 ms
64 bytes from 10.10.5.2: icmp_seq=4 ttl=64 time=0.269 ms

--- 10.10.5.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.269/0.293/0.318/0.021 ms
ceadmin@ceadmin-HP-Compaq-Pro-6300-MT:~$ ping -c 4 10.10.4.1
PING 10.10.4.1 (10.10.4.1) 56(84) bytes of data.
64 bytes from 10.10.4.1: icmp_seq=1 ttl=64 time=0.932 ms
64 bytes from 10.10.4.1: icmp_seq=2 ttl=64 time=0.928 ms
64 bytes from 10.10.4.1: icmp_seq=3 ttl=64 time=1.00 ms
64 bytes from 10.10.4.1: icmp_seq=4 ttl=64 time=0.977 ms

--- 10.10.4.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 0.928/0.959/1.000/0.030 ms

```

Figure 14. Direct-link ping test on PC F.

Clockwise Routing: A to D via B and C

The destination address selected on PC D for the clockwise test was 10.10.2.2 (D's interface on the C–D link), which naturally directed traffic along the A → B → C → D path. The routes installed were:

Forward path (A to D):

```

# On A:  sudo ip route replace 10.10.2.0/24 via 10.10.0.2
# On B:  sudo ip route replace 10.10.2.0/24 via 10.10.1.2

```

Return path (D to A):

```

# On D:  sudo ip route replace 10.10.0.0/24 via 10.10.2.1
# On C:  sudo ip route replace 10.10.0.0/24 via 10.10.1.1

```

PC C required no additional route for the forward direction because 10.10.2.0/24 was directly connected.

```

ceadmin@ce-558:~$ ping -c 4 10.10.2.2
PING 10.10.2.2 (10.10.2.2) 56(84) bytes of data.
64 bytes from 10.10.2.2: icmp_seq=1 ttl=62 time=0.720 ms
64 bytes from 10.10.2.2: icmp_seq=2 ttl=62 time=0.686 ms
64 bytes from 10.10.2.2: icmp_seq=3 ttl=62 time=0.729 ms
64 bytes from 10.10.2.2: icmp_seq=4 ttl=62 time=0.686 ms

--- 10.10.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3080ms
rtt min/avg/max/mdev = 0.686/0.705/0.729/0.019 ms
ceadmin@ce-558:~$ ip route get 10.10.2.2
10.10.2.2 via 10.10.0.2 dev enp2s0 src 10.10.0.1 uid 1000
cache
ceadmin@ce-558:~$ ping -c 4 10.10.2.2
PING 10.10.2.2 (10.10.2.2) 56(84) bytes of data.
64 bytes from 10.10.2.2: icmp_seq=1 ttl=62 time=0.743 ms
64 bytes from 10.10.2.2: icmp_seq=2 ttl=62 time=0.640 ms
64 bytes from 10.10.2.2: icmp_seq=3 ttl=62 time=0.703 ms
64 bytes from 10.10.2.2: icmp_seq=4 ttl=62 time=0.731 ms

--- 10.10.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.640/0.704/0.743/0.039 ms

```

Figure 15. Route table and clockwise ping result on PC A.

```

ceadmin@ce-558:~$ ip route get 10.10.2.2
10.10.2.2 via 10.10.0.2 dev enp2s0 src 10.10.0.1 uid 1000
cache

```

Figure 16. Confirmed reachability: A to D clockwise

Anti-Clockwise Routing: A to D via F and E

For the anti-clockwise test, the destination address on PC D was changed to 10.10.3.1 (D's interface on the D–E link), directing traffic along $A \rightarrow F \rightarrow E \rightarrow D$. Routes installed:

Forward path (A to D):

```

# On A:  sudo ip route replace 10.10.3.0/24 via 10.10.5.1
# On F:  sudo ip route replace 10.10.3.0/24 via 10.10.4.1

```

Return path (D to A):

```

# On D:  sudo ip route replace 10.10.5.0/24 via 10.10.3.2
# On E:  sudo ip route replace 10.10.5.0/24 via 10.10.4.2

```

```

ceadmin@ce-558:~$ ping -c 4 10.10.3.1
PING 10.10.3.1 (10.10.3.1) 56(84) bytes of data.
64 bytes from 10.10.3.1: icmp_seq=1 ttl=62 time=1.43 ms
64 bytes from 10.10.3.1: icmp_seq=2 ttl=62 time=1.43 ms
64 bytes from 10.10.3.1: icmp_seq=3 ttl=62 time=1.36 ms
64 bytes from 10.10.3.1: icmp_seq=4 ttl=62 time=1.51 ms

--- 10.10.3.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.357/1.429/1.505/0.052 ms
ceadmin@ce-558:~$ ip route get 10.10.3.1
10.10.3.1 via 10.10.5.1 dev enp3s0 src 10.10.5.2 uid 1000
    cache

```

Figure 17. Confirmed reachability: A to D anti-clockwise

Results

Test	Path	Result
A → B (direct)	—	Success
A → F (direct)	—	Success
B → C (direct)	—	Success
C → D (direct)	—	Success
D → E (direct)	—	Success
E → F (direct)	—	Success
A → D clockwise	A→B→C→D	Success
D → A clockwise return	D→C→B→A	Success
A → D anti-clockwise	A→F→E→D	Success
D → A anti-clockwise return	D→E→F→A	Success